Ashmit Dutta

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Education

University of Illinois Urbana-Champaign Urbana, IL
Bachelor of Science, Computer Engineering *Aug 2023 – May 2027*

- Co-author on an ICML 2025 workshop paper on finetuning Chain of Thought in LLMs.
- Built a 2-way superscalar out-of-order CPU (ECE 411, top 10) and a Unix-like RISC-V OS (ECE 391) from scratch.

Experience

Nablon AI Jun 2026 – Present
AI Engineering Intern *Remote — San Francisco, CA*

- Building **Sentinel**, an automated LLM-evaluation platform for an enterprise client: a **5-tier scanner pipeline** (rules, embeddings, fine-tuned classifiers, SLMs, LLM-as-judge) flagging jailbreaks, PII, toxicity, and hallucinations.
- Served it via **FastAPI** and an async Python worker pool with Langfuse / OpenTelemetry tracing on Azure.

Assured and Trustworthy AI Research Lab – University of Illinois Dec 2025 – Present
ML Research Intern (Prof. Huan Zhang) *Urbana, IL*

- Implemented the **Softmax** operator in **auto_LiRPA**, an open-source neural-network verification library (VNN-COMP winner); wrote C++ patches **cutting peak GPU memory 44%** and Python sanity-checking pipelines across 16+ benchmarks (ResNets, ViTs, YOLO, control nets).

John Deere Jan 2025 – Jan 2026
Software Engineer Intern *Champaign, IL*

- Shipped a React + Spring Boot platform **to production** for **500+ users**, built on REST APIs and CI/CD over PostgreSQL/DB2 handling **millions of records**.
- Designed a multithreaded job scheduler **cutting DB search time 3x**; added AWS CloudWatch observability.

University of Chicago Data Science Institute — DSI Summer Lab Jun 2024 – Aug 2024
ML Research Intern (Fermilab collaboration) *Chicago, IL*

- Built a reusable multi-modal GNN library (PyTorch Geometric) and an end-to-end training pipeline over **20,000+ detector events** on a Slurm GPU cluster for Fermilab's Exa.TrkX project.

Projects

Multi-Modal Video Search Engine | *Python, Modal, FastAPI* Feb 2026

- **Winner — Modal Inference + Bright Data AI tracks at Stanford TreeHacks 2026** (out of 380 teams).
- Built a distributed full-stack video search and dataset-generation app on **Modal** cloud GPU inference.
- Designed a multi-modal embedding pipeline fusing vision and audio for semantic retrieval, indexing 24 hr+ of video.

GPU-Accelerated Deep Learning Inference (LeNet-5) | *C++, CUDA* Feb 2026 – Apr 2026

- Wrote shape-specialized implicit-GEMM CUDA kernels and a WMMA Tensor Core conv for a **6.6x speedup** (12.5 ms inference at batch 10,000); drove design via Nsight Compute, **placing 5/130 in the ECE 408 competition**.

Unix-like RISC-V Operating System | *C, RISC-V, QEMU* Mar 2025 – May 2025

- Built a Unix-like OS from scratch with a **trap-based preemptive scheduler**, Sv39 virtual memory, and UART/VIRTIO/RTC device drivers.
- Implemented syscalls, ELF program loading, and a read/write filesystem, debugged with GDB/QEMU.

Leadership, Publications & Awards

Co-Founder & Lead Organizer, Online Physics Olympiad (OPhO): founded the largest student-run international physics competition (20k+ community); led website development and a team of 20+ students.

Awards & Leadership:
- *2nd Place, Bradley Mottier Innovation Challenge*, UIUC ISE ($2,000) — AR glasses for ASL translation.
- *Executive Board Member (Treasurer)*, IEEE Chapter at UIUC.
- Undergraduate Teaching Assistant, CS 450 (Numerical Analysis) — Publications: Google Scholar

Technical Skills

Languages: Python, C/C++, CUDA, Java, JavaScript, SQL
ML & GPU: PyTorch/Geometric, CUDA, GPU programming, inference optimization, model serving, model evaluation
Systems & Performance: low-latency C++20, lock-free concurrency, distributed inference, ML infrastructure, Nsight Compute, Slurm HPC
Infra: Docker, AWS, Modal, FastAPI, REST APIs, PostgreSQL, MongoDB, CI/CD, Git, Linux